GOOGLE CLOUD PROFESSIONAL CLOUD ARCHITECT

Section 2.2 — Planning and Configuring Data Storage Options

GCP Associate Cloud Engineer

Study Notes Study Notes April 2026

Exam Relevance

This topic is part of **Section 2: Planning and configuring a cloud solution (~17.5 % of the exam)**. You must know how to choose the right database or storage product for a given scenario and understand storage class differences.

1. Database and Storage Decision Overview

Google Cloud offers multiple storage and database services, each optimized for specific use cases:

```
                    Relational                     Non-Relational
                    ──────────                     ──────────────
Transactional       Cloud SQL                      Firestore
                    AlloyDB                        Bigtable
                    Spanner

Analytical          BigQuery                       BigQuery
                    AlloyDB (OLAP)

Object Storage      Cloud Storage (Standard, Nearline, Coldline, Archive)
Block Storage       Persistent Disk (Zonal, Regional)
File Storage        Filestore
```

2. Cloud SQL

What Is Cloud SQL?

- **Fully managed relational database** service
- Supports **MySQL**, **PostgreSQL**, and **SQL Server**
- Handles backups, replication, patches, and failover automatically

Key Features

- Up to **128 vCPUs** and **864 GB RAM**
- Up to **64 TB storage**
- Automatic storage increase
- High availability with **regional failover** (synchronous replication)
- Read replicas (within region, cross-region, and external)
- Point-in-time recovery (PITR)
- Private IP and public IP connectivity
- Cloud SQL Auth Proxy for secure connections

When to Choose Cloud SQL

- Traditional relational workloads (web apps, CMS, ERP)
- Existing MySQL/PostgreSQL/SQL Server applications
- Need for ACID transactions
- Data size up to ~64 TB
- Moderate throughput requirements

When NOT to Choose Cloud SQL

- Need horizontal scaling beyond a single region
- Data exceeds 64 TB
- Need global distribution with strong consistency
- Need millisecond-latency at massive scale
- Analytical workloads requiring petabyte-scale processing

Pricing

- Per vCPU, memory, and storage (per second, 1-minute minimum)
- Network egress charges
- HA instances cost ~2x (two instances running)

3. AlloyDB

What Is AlloyDB?

- **Fully managed, PostgreSQL-compatible** database
- Combines the best of open-source PostgreSQL with Google's infrastructure
- Up to **4x faster for transactional** and **100x faster for analytical** queries than standard PostgreSQL

Key Features

- PostgreSQL 14+ compatible
- Columnar engine for analytical queries
- ML-driven adaptive caching and indexing
- 99.99% SLA with regional availability
- Automated backups and PITR

When to Choose AlloyDB

- High-performance PostgreSQL workloads
- Mixed transactional and analytical (HTAP) workloads
- Migrating from Oracle or other enterprise databases
- Need PostgreSQL compatibility with better performance than Cloud SQL

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4. Cloud Spanner

What Is Cloud Spanner?

- **Globally distributed, strongly consistent** relational database
- Combines relational structure with horizontal scalability
- **Unlimited scale** with 99.999% SLA (five nines)

Key Features

- SQL interface with relational schemas
- ACID transactions across regions and continents
- Horizontal scaling (add nodes for more throughput)
- Automatic sharding
- Multi-region and single-region configurations

- Change streams for real-time data processing

When to Choose Cloud Spanner

- **Global applications** needing strong consistency (financial systems, inventory management)
- Relational data that **exceeds single-node capacity** (>64 TB or high throughput)
- Applications requiring **99.999% availability**
- Gaming leaderboards, ad-tech, retail supply chain

When NOT to Choose Cloud Spanner

- Small-scale applications (Cloud SQL is more cost-effective)
- Non-relational data
- Budget-constrained projects (Spanner starts at ~$0.90/node/hour)
- Workloads that can tolerate eventual consistency

Pricing

- Per node-hour (minimum 1 node)
- Per GB of storage
- Multi-region configurations cost more than single-region

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5. BigQuery

What Is BigQuery?

- **Serverless, highly scalable data warehouse**
- Designed for **analytical workloads** (OLAP)
- Processes petabytes of data using SQL
- Separate storage and compute (pay independently)

Key Features

- Serverless — no infrastructure management
- Standard SQL support
- Real-time analytics with streaming inserts
- ML built-in (BigQuery ML)
- Geospatial analysis (BigQuery GIS)
- BI Engine for fast dashboards
- Automatic data encryption

- Column-oriented storage (Capacitor format)
- Slots-based pricing for predictable costs (optional)

When to Choose BigQuery

- **Data warehousing** and business intelligence
- **Ad-hoc SQL queries** on large datasets
- **Log analytics** and clickstream analysis
- **Machine learning** with SQL (BigQuery ML)
- **Real-time dashboards** and reporting
- Processing data from Cloud Storage, Cloud SQL, Sheets, etc.

When NOT to Choose BigQuery

- Transactional (OLTP) workloads (use Cloud SQL or Spanner)
- Low-latency point lookups (use Bigtable or Firestore)
- Streaming with sub-second requirements (use Bigtable)
- Small datasets that don't benefit from distributed processing

Pricing Models

- **On-demand**: $6.25 per TB processed (first 1 TB/month free)
- **Capacity (Editions)**: Purchase slots for predictable costs
- **Storage**: $0.02/GB/month (active), $0.01/GB/month (long-term, 90+ days untouched)
- **Streaming inserts**: $0.05 per GB

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6. Firestore (Datastore Mode and Native Mode)

What Is Firestore?

- **Serverless, NoSQL document database**
- Designed for mobile, web, and IoT applications
- Real-time sync and offline support (Native mode)
- Scalable to millions of concurrent connections

Two Modes

Feature	Native Mode	Datastore Mode
Data model	Documents in collections	Entities in kinds
Real-time listeners	Yes	No
Offline support	Yes (mobile SDKs)	No
Transactions	Multi-document ACID	Entity group transactions
Best for	Mobile/web apps	Server-side applications
Querying	Collection queries, collection group queries	GQL, structured queries

When to Choose Firestore

- **Mobile and web applications** needing real-time sync
- **User profiles**, session data, game state
- **Product catalogs** with hierarchical data
- **IoT device data** collection
- Semi-structured or hierarchical data

When NOT to Choose Firestore

- Analytical queries across millions of records (use BigQuery)
- Time-series data at massive scale (use Bigtable)
- Need for complex joins and relational queries (use Cloud SQL)
- Data exceeding 1 TB per database (consider Bigtable)

Pricing

- Per document read, write, and delete
- Per GB stored
- Free tier: 50K reads, 20K writes, 20K deletes per day

7. Cloud Bigtable

What Is Cloud Bigtable?

- **Fully managed, wide-column NoSQL database**
- Designed for **massive analytical and operational workloads**
- Same technology that powers Google Search, Maps, and Gmail

Key Features

- Millisecond latency at any scale
- Handles billions of rows and thousands of columns
- Scales linearly by adding nodes
- Native HBase API compatibility
- Integration with Hadoop, Dataflow, and Dataproc
- Automatic replication across zones and regions
- SSD or HDD storage options

When to Choose Cloud Bigtable

- **Time-series data** (IoT, financial ticks, monitoring metrics)
- **Marketing analytics** (user behavior, ad impressions)
- **Graph data** and relationships
- Workloads requiring **>1 TB of data** with low latency
- **MapReduce** or stream processing pipelines
- High-throughput **read/write** applications (>10,000 QPS)

When NOT to Choose Cloud Bigtable

- Data less than 1 TB (not cost-effective at small scale)
- Need for complex SQL queries (use BigQuery or Cloud SQL)
- ACID transactions across multiple rows (use Spanner)
- Document storage (use Firestore)

Pricing

- Per node-hour (minimum 1 node, recommended 3+ for production)
- Per GB of storage (SSD or HDD)
- Network egress

8. Database Comparison Matrix

Feature	Cloud SQL	AlloyDB	Spanner	BigQuery	Firestore	Bigtable
Type	Relational	Relational	Relational	Analytical	Document (NoSQL)	Wide-column (NoSQL)
Workload	OLTP	OLTP + OLAP	OLTP	OLAP	Mobile/Web	Time-series, IoT
Scale	Vertical	Vertical	Horizontal	Serverless	Serverless	Horizontal
Max data	64 TB	128+ TB	Unlimited	Unlimited	1 TB/database	Unlimited
Consistency	Strong	Strong	Strong (global)	Eventual	Strong	Eventual
SQL support	Full SQL	Full SQL (PG)	Google SQL	Standard SQL	No (queries)	No
SLA	99.95-99.99%	99.99%	99.999%	99.99%	99.999%	99.99%
Global	Read replicas	Read replicas	Multi-region	Multi-region	Multi-region	Multi-region
Min cost	Low	Medium	High	Free tier	Free tier	Medium

9. Cloud Storage Classes

Cloud Storage is Google's object storage service. Data is organized in **buckets** and stored as **objects** (files).

Storage Classes

Class	Min Storage Duration	Retrieval Cost	Use Case
Standard	None	Free	Frequently accessed data (“hot data”)
Nearline	30 days	Per GB	Data accessed less than once per month
Coldline	90 days	Per GB (higher)	Data accessed less than once per quarter
Archive	365 days	Per GB (highest)	Data accessed less than once per year

Key Pricing Concepts

- **Storage cost decreases** as you move from Standard → Archive
- **Retrieval cost increases** as you move from Standard → Archive
- **Early deletion charges**: If you delete data before the minimum storage duration, you’re charged for the remaining time
- All classes have the **same millisecond access latency** (unlike AWS Glacier)

Storage Class Comparison

```
                 Storage Cost    Retrieval Cost    Min Duration
Standard    ████████████████    ░░░░░░░░░░░░░░    None
Nearline    ██████████████░░    ██░░░░░░░░░░░░    30 days
Coldline    ████████████░░░░    ████░░░░░░░░░░    90 days
Archive     ██████████░░░░░░    ██████░░░░░░░░    365 days
```

Bucket-Level vs. Object-Level Storage Class

- **Default storage class** is set at the bucket level
- Individual objects can have **different storage classes** within the same bucket
- Use **Object Lifecycle Management** to automatically transition objects between classes

Common Use Cases

Storage Class	Example Use Cases
Standard	Website assets, streaming media, mobile app data, gaming content
Nearline	Monthly backups, long-tail multimedia, data for monthly analytics
Coldline	Disaster recovery, quarterly accessed data
Archive	Regulatory compliance archives, long-term backups, digital preservation

10. Persistent Disk Options

Persistent Disks are durable block storage for Compute Engine and GKE.

Disk Types

Type	Description	IOPS (Read/Write)	Throughput	Use Case
pd-standard	Standard HDD	Low	Low	Bulk storage, sequential I/O
pd-balanced	Balanced SSD	Medium	Medium	General-purpose (default)
pd-ssd	Performance SSD	High	High	Databases, random I/O
pd-extreme	Highest performance SSD	Highest	Highest	Mission-critical databases

Zonal vs. Regional Persistent Disks

Feature	Zonal PD	Regional PD
Availability	Single zone	Two zones in same region
Replication	Within zone	Synchronous across 2 zones
Use case	Standard workloads	HA workloads needing failover
Cost	Standard	~2x zonal cost
Failover	Manual	Faster (data already in 2 zones)

Local SSDs

- Physically attached to the host machine
- **Extremely high IOPS** (up to 2.4M read IOPS)
- **Ephemeral** — data lost when VM stops or is deleted
- Use cases: Caches, temp storage, scratch processing
- Fixed sizes: 375 GB per device (up to 24 devices)

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11. Filestore

What Is Filestore?

- **Managed NFS file storage** service
- Provides shared file system accessible by multiple VMs
- Use cases: Media rendering, EDA, genome processing, web content

Tiers

- **Basic HDD / Basic SSD** — Development and testing
- **High Scale SSD** — High-performance workloads
- **Enterprise** — Mission-critical, regional availability

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12. Caching — Memorystore

- **Memorystore** is fully managed Redis or Memcached

- Use when: repeated reads of the same data, session storage, rate limiting, leaderboards
- **Redis**: supports persistence, replication (STANDARD_HA tier), Lua scripting, Pub/Sub, Streams
- **Memcached**: multi-threaded, simpler, pure cache (no persistence)
- Decision: default to Redis; use Memcached only when multi-threading is required and no persistence needed
- Planning tip: Size the cache to fit your working set; typical starting point is 20-30% of dataset size

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13. BigQuery Partitioning and Clustering

- **Partitioning**: divides a table into segments by a column value; queries that filter on the partition column scan only relevant partitions (reduces cost and latency)
  - Partition types: ingestion time (auto), DATE/TIMESTAMP/DATETIME column, INTEGER range
  - Max 4,000 partitions per table
- **Clustering**: sorts data within each partition by up to 4 columns; further reduces bytes scanned for queries that filter on clustered columns
  - Clustering columns: high-cardinality columns used in WHERE or GROUP BY
  - Clusters are automatically maintained as data is inserted

Planning Guidance

If you filter by...	Use...
Date/time	Date partitioning
Integer ranges	Integer range partitioning
High-cardinality label	Clustering
Both	Partition + cluster

- Exam tip: Partitioning reduces cost (fewer bytes billed); clustering reduces latency (less data scanned within partition). Both are free features.

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14. Encryption Options

- **Default**: all GCP data encrypted at rest with Google-managed keys (AES-256) — automatic, no configuration needed
- **CMEK (Customer-Managed Encryption Keys)**: you manage keys in Cloud KMS; GCP uses them to encrypt your data
  - Use case: compliance requirements, key lifecycle control, ability to revoke access by disabling the key
  - Supported by: Cloud Storage, BigQuery, Compute Engine disks, Cloud SQL, Pub/Sub, Cloud Run, GKE

```
# Create a KMS key ring and key
gcloud kms keyrings create my-keyring --location=us-central1
gcloud kms keys create my-key --keyring=my-keyring --location=us-central1 --purpose=enc
# Apply to a GCS bucket
gcloud storage buckets create gs://my-bucket --default-encryption-key=projects/PROJECT/
```

- **CSEK (Customer-Supplied Encryption Keys)**: you supply raw key material to GCP; GCP never stores it; you must supply the key for every operation; highest control, highest responsibility
  - Only supported by Cloud Storage and Compute Engine disks
- Exam tip: CMEK = keys in Cloud KMS (GCP stores metadata, you own the key); CSEK = raw key you supply per-request (GCP stores nothing); Google-managed = simplest, no action required

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Exam Practice Questions

1. **A global financial services company needs a relational database with 99.999% availability and strong consistency across continents. Which service should they use?**
   - Answer: **Cloud Spanner** — Only GCP database offering global strong consistency with 99.999% SLA.
2. **You need to store and analyze 5 PB of historical log data using SQL. Which service should you use?**
   - Answer: **BigQuery** — Serverless data warehouse designed for petabyte-scale SQL analytics.

3. **A mobile app needs real-time sync of user data across devices with offline support. Which database should you use?**
   - Answer: **Firestore (Native mode)** — Built for mobile/web with real-time listeners and offline support.
4. **You need to store backup data that is accessed about once per quarter. Which storage class minimizes cost?**
   - Answer: **Coldline** — Designed for data accessed less than once per quarter, with lower storage costs than Standard/Nearline.
5. **An IoT application generates millions of time-series data points per second with low-latency read requirements. Which database is best?**
   - Answer: **Cloud Bigtable** — Designed for high-throughput time-series data with millisecond latency.
6. **A web application uses PostgreSQL and needs to handle both transactional queries and complex analytical reports. Which service would you recommend?**
   - Answer: **AlloyDB** — PostgreSQL-compatible with excellent performance for both OLTP and OLAP (HTAP) workloads.
7. **What is the key difference between Nearline and Coldline storage?**
   - Answer: Minimum storage duration (30 days vs 90 days) and retrieval costs (Coldline is more expensive to retrieve). Both have the same access latency (milliseconds).
8. **You need a block storage device that provides high availability across zones for a production database. What should you use?**
   - Answer: **Regional Persistent Disk (pd-ssd or pd-extreme)** — Synchronously replicates data across two zones for HA.

Glossary

ACID (Atomicity, Consistency, Isolation, Durability) — A set of properties that guarantee database transactions are processed reliably; Cloud SQL, AlloyDB, and Spanner provide full ACID compliance.

AlloyDB — GCP's fully managed, PostgreSQL-compatible database service combining open-source PostgreSQL with Google's infrastructure; up to 4x faster for OLTP and 100x faster for analytical queries than standard PostgreSQL.

Archive — The coldest Cloud Storage class, with a minimum storage duration of 365 days; lowest storage cost but highest retrieval cost; designed for data accessed less than once per year.

Automatic Sharding — A feature of Cloud Spanner that automatically distributes data across nodes based on key ranges without manual partitioning, enabling horizontal scalability.

AES-256 (Advanced Encryption Standard, 256-bit) — The encryption algorithm used by GCP to protect all data at rest by default using Google-managed keys.

BI Engine — BigQuery's in-memory analysis service that accelerates SQL queries for business intelligence dashboards and reporting tools.

BigQuery — GCP's serverless, highly scalable data warehouse designed for OLAP analytical workloads; processes petabytes of data using Standard SQL with separate storage and compute billing.

BigQuery GIS — BigQuery feature for geospatial analysis, enabling SQL queries over geographic data types such as points, lines, and polygons.

BigQuery ML — Feature within BigQuery that enables creating, training, and executing machine learning models using SQL statements without moving data.

Bigtable (Cloud Bigtable) — GCP's fully managed wide-column NoSQL database built for massive analytical and operational workloads; powers Google Search, Maps, and Gmail internally; offers millisecond latency at any scale.

Block Storage — Storage type that presents raw storage volumes to VMs as block devices; Persistent Disk and Local SSD are GCP's block storage options.

Bucket — The fundamental container in Cloud Storage where objects (files) are stored; every object belongs to exactly one bucket.

Capacitor — The columnar storage format used internally by BigQuery to optimize compression and scan performance for analytical queries.

Change Streams — A Cloud Spanner feature that captures data changes in real time, enabling event-driven architectures and downstream data synchronization.

Cloud KMS (Key Management Service) — GCP's managed service for creating, rotating, and controlling cryptographic keys; used with CMEK to encrypt data across multiple GCP services.

Cloud SQL — GCP's fully managed relational database service supporting MySQL, PostgreSQL, and SQL Server; handles backups, replication, patches, and failover automatically.

Cloud SQL Auth Proxy — A binary that provides secure, authenticated access to Cloud SQL instances without requiring SSL certificates or authorized networks.

Cloud Storage — GCP's object storage service for storing and retrieving unstructured data in buckets; offers four storage classes (Standard, Nearline, Coldline, Archive) with millisecond access latency for all classes.

CMEK (Customer-Managed Encryption Keys) — An encryption option where the customer manages keys in Cloud KMS and GCP uses those keys to encrypt data; allows key revocation and lifecycle control.

Coldline — A Cloud Storage class designed for data accessed less than once per quarter; has a 90-day minimum storage duration with higher retrieval costs than Standard or Nearline.

Collection — In Firestore Native mode, a grouping of documents analogous to a table in relational databases; collections can contain sub-collections for hierarchical data.

Columnar Engine — AlloyDB's built-in analytical acceleration layer that stores data in column-oriented format for fast analytical query execution alongside transactional workloads.

CSEK (Customer-Supplied Encryption Keys) — An encryption option where the customer supplies raw key material directly to GCP per request; GCP never stores the key; supported only by Cloud Storage and Compute Engine disks.

Dataflow — GCP's fully managed stream and batch data processing service based on the Apache Beam programming model; integrates natively with Bigtable, BigQuery, and Cloud Storage.

Dataproc — GCP's managed Apache Hadoop and Spark service; integrates with Bigtable for MapReduce workloads.

Datastore Mode — A Firestore operational mode backward-compatible with the original Cloud Datastore API; uses entities and kinds instead of documents and collections; no real-time listeners or offline support.

Document — The fundamental data unit in Firestore; a set of key-value pairs (fields) organized within a collection; analogous to a row in a relational database.

Early Deletion Charge — A Cloud Storage fee applied when an object is deleted before its storage class's minimum storage duration expires; the charge covers the remaining minimum duration period.

EDA (Electronic Design Automation) — Engineering workloads for designing integrated circuits; a use case for Filestore's shared file storage.

Encryption at Rest — Encryption of data stored on physical media; GCP encrypts all data at rest by default using AES-256 with Google-managed keys.

Entity — The data record type in Firestore Datastore mode; equivalent to a document in Native mode; belongs to a "kind" (analogous to a collection).

Eventual Consistency — A consistency model in which, given enough time without updates, all replicas will converge to the same value; used by Bigtable and BigQuery; contrasts with strong consistency.

File Storage — Storage type providing a shared file system accessible by multiple clients via standard file protocols; Filestore is GCP's managed file storage service.

Filestore — GCP's managed NFS file storage service providing shared file systems accessible by multiple Compute Engine VMs and GKE pods; available in Basic HDD, Basic SSD, High Scale SSD, and Enterprise tiers.

Firestore — GCP's serverless NoSQL document database designed for mobile, web, and IoT applications; available in Native mode (real-time sync, offline support) and Datastore mode (server-side compatibility).

Five Nines (99.999% SLA) — The highest availability tier offered by GCP, provided by Cloud Spanner and Firestore; allows less than 5.26 minutes of downtime per year.

GQL (Google Query Language) — SQL-like query language used in Firestore Datastore mode for querying entities.

HA (High Availability) — Design approach ensuring a system remains operational with minimal downtime; achieved in Cloud SQL via synchronous regional failover, and in Persistent Disk via Regional PD.

HBase API — The Apache HBase client API; Cloud Bigtable is compatible with it, allowing Hadoop ecosystem tools to read from and write to Bigtable without code changes.

HDD (Hard Disk Drive) — Magnetic spinning disk storage; used in pd-standard Persistent Disks and Filestore Basic HDD tier; lower cost but lower IOPS than SSD.

Hot Data — Frequently accessed data; best stored in Cloud Storage Standard class or in databases with low-latency access patterns.

HTAP (Hybrid Transactional/Analytical Processing) — A database workload pattern combining OLTP and OLAP queries; AlloyDB is optimized for HTAP workloads.

IAM (Identity and Access Management) — GCP's system for controlling access to resources; applies to Cloud Storage buckets, BigQuery datasets, and all other storage services.

Integer Range Partitioning — A BigQuery partitioning strategy that divides a table into segments based on integer column value ranges.

IOPS (Input/Output Operations Per Second) — A measure of storage performance; pd-ssd and pd-extreme Persistent Disks offer high IOPS for database workloads.

IoT (Internet of Things) — Network of physical devices generating sensor data; a primary use case for Firestore (device state) and Bigtable (time-series metrics).

Kind — In Firestore Datastore mode, the category or type of an entity; analogous to a table in relational databases or a collection in Native mode.

Local SSD — Physically attached NVMe SSD on a Compute Engine host; offers up to 2.4 million read IOPS but is ephemeral — data is lost when the VM stops or is deleted; sold in 375 GB increments.

Long-Term Storage Pricing — BigQuery's reduced storage pricing ($0.01/GB/month) automatically applied to tables or partitions not modified for 90 or more consecutive days.

MapReduce — Distributed data processing programming model; Bigtable integrates natively with MapReduce via its HBase API compatibility.

Memcached — Open-source, multi-threaded in-memory key-value caching system; available via GCP Memorystore; suitable when no persistence or complex data structures are needed.

Memorystore — GCP's fully managed in-memory caching service supporting Redis and Memcached; used for session storage, rate limiting, leaderboards, and frequently read data.

Multi-Region — A Cloud Storage or database configuration spanning multiple geographic regions for higher availability and lower latency for geographically distributed users.

MySQL — Open-source relational database management system; one of the three database engines supported by Cloud SQL.

Native Mode — The default and recommended Firestore operational mode; supports real-time listeners, offline sync for mobile SDKs, multi-document ACID transactions, and collection group queries.

Nearline — A Cloud Storage class designed for data accessed less than once per month; has a 30-day minimum storage duration with lower retrieval costs than Coldline or Archive.

NFS (Network File System) — Standard file-sharing protocol; used by Filestore to provide shared file access to multiple Compute Engine VMs and GKE pods.

NoSQL — A category of database management systems that do not use traditional relational (tabular) data models; Firestore and Bigtable are GCP's primary NoSQL databases.

Object — The fundamental unit stored in Cloud Storage; consists of data (the file content) plus metadata; objects are stored within buckets.

Object Lifecycle Management — A Cloud Storage feature that automatically transitions objects between storage classes or deletes them based on configurable rules (age, storage class, number of versions, etc.).

OLAP (Online Analytical Processing) — Workload category focused on complex analytical queries over large datasets; BigQuery and AlloyDB's columnar engine serve OLAP workloads.

OLTP (Online Transactional Processing) — Workload category focused on fast, high-volume read/write transactions; Cloud SQL, AlloyDB, and Spanner serve OLTP workloads.

On-Demand Pricing (BigQuery) — BigQuery pricing model charging $6.25 per TB of data processed by queries; the first 1 TB per month is free.

Oracle — Third-party enterprise relational database; AlloyDB is commonly chosen as a migration target for Oracle workloads due to its PostgreSQL compatibility and high performance.

Partitioning (BigQuery) — A BigQuery optimization that divides a table into segments by a column value (date/time, integer range, or ingestion time); queries filtering on the partition column scan only relevant partitions, reducing cost.

pd-balanced — Balanced SSD Persistent Disk type offering medium IOPS and throughput; the default general-purpose disk type for Compute Engine.

pd-extreme — Highest-performance SSD Persistent Disk type providing the maximum IOPS and throughput; designed for mission-critical database workloads.

pd-ssd — Performance SSD Persistent Disk type offering high IOPS and throughput; suitable for database workloads requiring fast random I/O.

pd-standard — Standard HDD Persistent Disk type with low IOPS; suited for bulk storage and sequential I/O workloads where cost is prioritized over performance.

Persistent Disk — GCP's durable network-attached block storage for Compute Engine VMs and GKE nodes; available as Zonal or Regional PD in HDD and SSD variants.

PITR (Point-in-Time Recovery) — Database feature that allows restoring data to any specific moment within a retention window; supported by Cloud SQL and AlloyDB.

PostgreSQL — Open-source object-relational database management system; one of the three engines supported by Cloud SQL and the compatibility layer for AlloyDB and Spanner.

QPS (Queries Per Second) — A measure of read/write throughput; Bigtable is recommended for workloads exceeding 10,000 QPS with low-latency requirements.

Read Replica — A copy of a database that serves read traffic, offloading it from the primary instance; supported by Cloud SQL within a region, cross-region, and externally.

Redis — Open-source in-memory data structure store supporting persistence, replication, Pub/Sub, and Lua scripting; available via GCP Memorystore; the default recommendation for caching.

Regional Persistent Disk — A Persistent Disk that synchronously replicates data across two zones within a single region; provides faster failover and higher availability than Zonal PD at approximately 2x the cost.

Relational Database — A database that organizes data into structured tables with defined schemas and supports SQL queries; Cloud SQL, AlloyDB, and Spanner are GCP's relational database offerings.

Slots (BigQuery) — Units of BigQuery query processing capacity; purchased in the Capacity Editions pricing model for predictable, flat-rate query costs independent of data scanned.

Spanner (Cloud Spanner) — GCP's globally distributed, strongly consistent, horizontally scalable relational database; offers unlimited scale with a 99.999% SLA; supports ACID transactions across regions.

SQL (Structured Query Language) — Standard language for querying and manipulating relational data; supported by Cloud SQL, AlloyDB, Spanner (Google SQL dialect), and BigQuery (Standard SQL).

SQL Server — Microsoft's relational database management system; one of the three database engines supported by Cloud SQL.

SSD (Solid-State Drive) — Flash-based storage offering high IOPS and low latency; used in pd-balanced, pd-ssd, pd-extreme Persistent Disks, Local SSDs, and Bigtable SSD storage.

Standard (Storage Class) — The default Cloud Storage class with no minimum storage duration and no retrieval cost; designed for frequently accessed ("hot") data.

Standard SQL — ANSI-compliant SQL dialect used by BigQuery for querying data warehouse tables.

Streaming Inserts — A BigQuery feature for inserting individual rows in near-real time (rather than batch loads); enables real-time analytics at a per-GB cost.

Strong Consistency — A consistency model guaranteeing that all reads reflect the most recent write; provided by Cloud SQL, AlloyDB, Spanner (globally), and Firestore Native mode.

Sub-Collection — A collection nested within a Firestore document, enabling hierarchical data modeling without denormalization.

Throughput — The rate of data transfer measured in MB/s; a key performance characteristic of Persistent Disk types alongside IOPS.

Time-Series Data — Measurements recorded at successive points in time (e.g., IoT sensor readings, financial ticks, monitoring metrics); the primary use case for Cloud Bigtable.

Uniform Bucket-Level Access — A Cloud Storage setting (enforceable via org policy) that disables per-object ACLs and requires all access to be managed through IAM; simplifies access control.

Vertex AI — GCP's unified machine learning platform; relevant to data storage planning as it integrates with BigQuery for ML model training on structured data.

Wide-Column Database — A NoSQL database type that organizes data into rows with a flexible set of columns; Bigtable is GCP's wide-column database, similar in design to Apache HBase and Cassandra.

Zonal Persistent Disk — A Persistent Disk that resides within a single zone; the standard, lower-cost block storage option for Compute Engine when cross-zone HA is not required.